

You are the network controller.

A synthetic stadium surge is approaching. Vote before the result is revealed.

JOIN JUPYTERHUB

**Open your assigned notebook.
Run Stage 01 preflight.**

Reveal at minute 72.

A

Keep Static

B

React after overload

C

Steer predictively

What this digital twin represents

02 · 90-MINUTE WORKSHOP

REPRESENTS

Synthetic traffic, topology, capacity, telemetry, forecast uncertainty, optimization, policy gates and later evidence.

DOES NOT REPRESENT

Live C-DOT traffic, measured geography, autonomous production control, or established-session migration.

CONTROL SCOPE

Recommendations change only the placement of sessions that arrive after an accepted policy epoch.

Implemented here; external integration remains gated

03 · 90-MINUTE WORKSHOP

IMPLEMENTED

24-UPF/96-group synthetic model · deterministic simulator · HiGHS · policy gate · PBS campaign jobs · dashboard · replay.

CLUSTER-PREFLIGHT

SCIP, PySCIPOpt, ParaSCIP/UG, MPI/PALS and PBS launch syntax must pass on the actual cluster ≥ 7 days before delivery.

EXTERNAL - PENDING

C-DOT telemetry mapping · calibrated envelopes · security · SMF/EMS future-session hook · publication/rollback runbook.

One causal control loop—from pressure to later evidence

04 · 90-MINUTE WORKSHOP

OBSERVE → FORECAST

Offered demand and quality flags enter a closed-history p50/p90 forecast. Target data never enters features.

OPTIMIZE → CERTIFY

Capacity, eligibility, health, locality and churn gates evaluate a candidate. Unsafe means last-safe/static.

PLACE → MEASURE

Only future sessions use new weights. Carried traffic, overload and loss become later evidence—not rewritten history.

A synthetic national topology creates repeatable stress

05 · 90-MINUTE WORKSHOP

24 UPFs

Heterogeneous UL/DL/session capacity, safe envelopes, health and failure-domain events.

96 GROUPS

Zone × DNN × S-NSSAI demand classes with eligibility and deterministic offered-load generation.

TRAFFIC SEMANTICS

Offered demand is attempted load. Carried is an outcome. Their constrained gap is loss/overload.

The 160-node cluster buys breadth—not one giant simulation

06 · 90-MINUTE WORKSHOP

RIGHT UNIT

One scenario × controller × seed is one bounded one-node, one-CPU PBS job.

CAMPAIGN BREADTH

Independent jobs explore seeds, events, controllers and parameter regimes concurrently across the cluster.

ISOLATION

\$USER + PBS job ID + seed identify private outputs. No shared writable participant campaigns.

The teaching LP turns demand into safe routing weights

07 · 90-MINUTE WORKSHOP

VARIABLES

x_u = new-session Mbps assigned to eligible UPF u · s_u = explicit overload slack.

CONSTRAINTS

$\sum x_u = \text{demand}$ · $x_u \leq \text{residual capacity}_u + s_u$ · $x_u = 0$ when ineligible.

OBJECTIVE + OUTPUT

Minimize latency-weighted allocation + severe slack penalty.
Normalize $w_u = x_u / \text{demand}$.

SCIP validates the LP; ParaSCIP demonstrates the larger MIP

08 · 90-MINUTE WORKSHOP

INDIVIDUAL

Same LP in HiGHS and SCIP; feasible objectives must agree within tolerance. One-node SCIP job per participant.

WHY A MIP

24-UPF/96-group assignment uses binary group → UPF and UPF-activation variables. The tiny LP cannot teach distributed search.

PRESENTER ONLY

One frozen-seed, two-node ParaSCIP/UG job in a reservation. Participants watch status/result; they never submit ParaSCIP.

Offered demand and carried throughput answer different questions

09 · 90-MINUTE WORKSHOP

OFFERED

What active and newly arriving sessions attempted to transmit —independent of whether capacity served it.

CARRIED

What crossed the modeled service path after capacity, queueing, rejection and drops.

DO NOT CONFUSE

A saturated carried-throughput series can hide true demand. Forecast from offered demand with quality flags.

Forecast uncertainty changes what ‘safe’ means

1 0 · 9 0 - M I N U T E W O R K S H O P

CAUSAL INPUT

Only closed windows at or before issue time. Scheduled events enter only when known by then.

p50 vs p90

p50 is central expectation; p90 is a conservative planning quantile—not a guarantee against surprise flash crowds.

MARK EVERY FRAME

Source window, target window, policy apply step and anchored-session marker preserve replay semantics.

Certification makes rejection visible and fail-safe

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CHECK

Causality · finite normalized weights · eligibility · health · locality · UL/DL/session headroom · churn.

REJECT

Unknown destination, non-normalized weight, unsafe projection or migration request fails certification.

FALL BACK

Retain last-safe/static policy with reason, status, audit and expiry. Never publish slack or silently switch solvers.

The guided three-UPF story is positive within its matched scope

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EVIDENCE SCOPE · Frozen guided three-UPF showcase · 30 matched pairs · positive result

+10.52%

Matched 30-pair guided result in the frozen three-UPF showcase evidence. Present as a scoped modeled improvement.

WHAT CHANGED

Accepted weights changed future-session placement; established sessions stayed anchored.

WHAT IT DOES NOT PROVE

No live-network benefit, no universal overload prevention, no production recommendation.

The later national result does not promote MPC

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EVIDENCE SCOPE · Later national-scale control-science campaign · production decision

LATER EVIDENCE

Broader national-scale campaign evaluates the controller under more regimes and guardrails than the guided story.

DECISION

MPC fails promotion. Static remains the production-safe controller conclusion.

HONEST HANDOFF

The older +10.52% remains useful teaching evidence, but is not the latest production recommendation.

From dashboard and 3D replay to a bounded advisory pilot

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EXPERIENCE

Presenter dashboard streams the guided story; participant iframe replays completed Parquet in the full Three.js world.

OPERATIONAL GATES

35-user rehearsal · reserved short queue · offline assets · WebGL2 · storage pressure · 90-minute run + 5-minute margin.

C-DOT PATH

Map telemetry → calibrate envelopes → shadow/advisory replay → secure audited future-session pilot → rollback exercise.